

The displayed random-graph spectral-gap scale is unbounded

ziangni-sys Conjecture 00000007857

Disproof of the formula as written. Both language versions display

$$\gamma_N(p) = \Theta\left(\max\left(1 - \frac{\lambda_2(G)}{Np}, Np^{-1}\right)\right) \quad \text{when } p \gg \frac{\log N}{N}.$$

Here $Np^{-1} = N/p$ in standard notation. We disprove this formula, and make no assertion about a different formula obtained by replacing that term by $(Np)^{-1}$.

Take $p = 1$ and every integer $N \geq 2$. In the actual Erdős–Rényi model, the indicators of distinct unordered edges are independent Bernoulli random variables with parameter one. Each edge is present almost surely, so $G(N, 1) = K_N$ almost surely. This is also a direct one-point probability-space realization of the model: all edge indicators are the constant true variable, with their genuine Bernoulli(1) marginal laws and mutual independence. The stipulated dense regime holds, since

$$\frac{\log N}{N} \rightarrow 0.$$

The source imposes no restriction excluding $p = 1$.

Let A be the actual adjacency matrix of K_N . Every vertex has degree $d = N - 1$, and for every vector $x \in \mathbb{R}^N$,

$$(Ax)_i = \sum_{j=1}^N x_j - x_i.$$

The normalized Laplacian and lazy transition matrix are respectively

$$L = I - \frac{A}{N-1}, \quad P = \frac{1}{2}I + \frac{A}{2(N-1)}, \quad I - P = \frac{1}{2}L.$$

The entries of P are nonnegative and each row sums to one. Thus P is the actual lazy simple random-walk transition matrix, not a prescribed spectral surrogate.

Writing $\mathbf{1} = (1, \dots, 1)$ gives

$$Lx = \frac{N}{N-1}x - \frac{\sum_i x_i}{N-1}\mathbf{1}.$$

Therefore $L\mathbf{1} = 0$, and on the nonzero sum-zero subspace it acts by the scalar $N/(N-1)$. For completeness, if $cLx = rx$ with $x \neq 0$ and $r \neq 0$, summing the coordinates gives $r \sum_i x_i = 0$, hence $\sum_i x_i = 0$ and $r = cN/(N-1)$. Conversely, $e_1 - e_2$ is a genuine nonzero eigenvector for this scalar. For $c > 0$, the full positive eigenvalue set is consequently the singleton $\{cN/(N-1)\}$.

Taking the infimum of that actual positive eigenvalue set yields both conventions appearing in the source:

$$\gamma_N^{\text{norm}} = \frac{N}{N-1} \leq 2, \quad \gamma_N^{\text{lazy}} = \frac{N}{2(N-1)} \leq 1.$$

At $p = 1$, however, the asserted comparison scale satisfies

$$F_N = \max\left(1 - \frac{\lambda_2(K_N)}{N}, N\right) \geq N.$$

This inequality holds regardless of the value or ordering convention for λ_2 ; no spectral table for that term is required. If the asserted Θ lower bound held, there would be $a > 0$ and N_0 such that $aF_N \leq \gamma_N$ for all sufficiently large N . Choosing $N \geq N_0$ with $aN > 2$ contradicts either gap convention. The failure occurs with probability one, so a high-probability qualifier cannot repair it.

Formal correspondence. Lean uses actual complete `SimpleGraphs` and adjacency matrices, proves their degrees and matrix actions, and computes the full positive eigenvalue sets and their `sInf` gaps for arbitrary positive scaling. It verifies the lazy matrix's nonnegative entries and row sums. A one-point probability space supplies genuinely independent edge functions whose pushed-forward measures equal the actual `PMF.bernoulli 1` measure. The logarithmic dense-regime limit is proved, followed by failure of every positive eventual lower constant, uniformly for every possible second-eigenvalue sequence. The separate concentration-width and GOE assertions are not needed.